National University of Computer and Emerging Sciences

Lab Exercise 12
AL2002-Artifical Intelligence Lab

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Exercise

1. Use this dataset from kaggle: https://www.kaggle.com/datasets/hassnainzaidi/garbage-classification/data
    - Load the dataset from the directory structure using `image_dataset_from_directory` from tensorflow:

```
import tensorflow as tf

IMG_SIZE = (128, 128)
BATCH_SIZE = 32

train_dataset =
tf.keras.preprocessing.image_dataset_from_directory(
    "Garbage classification/train",
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE
)

val_dataset = tf.keras.preprocessing.image_dataset_from_directory(
    "Garbage classification/val",
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE
)

test_dataset =
tf.keras.preprocessing.image_dataset_from_directory(
    "Garbage classification/test",
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    shuffle=False   # Important for evaluation
)

class_names = train_dataset.class_names
print("Classes:", class_names)
```

2. Create a CNN model using TensorFlow/Keras with the following structure:
    - Input Layer
    - Rescaling (1./255)
    - Convolution Layer (32 filters, 3×3, ReLU)
    - MaxPooling Layer (2×2)

- Convolution Layer (64 filters, 3×3, ReLU)
    - MaxPooling Layer (2×2)
    - Flatten Layer
    - Dense Layer (64 neurons, ReLU)
    - Output Layer (Softmax, number of classes)

3. Use:
    - Optimizer: Adam
    - Loss Function: categorical_crossentropy (or sparse equivalent)
    - Metrics: accuracy

4. Train the model using:

    - epochs = 100

    - batch_size = 32

    - validation_split = 0.2

    - Apply Early Stopping with:

        - monitor = 'val_loss'

        - patience = 5

5. Plot the Training vs Validation Accuracy and Training vs Validation Loss to answer the following questions:

    - Does the model show signs of overfitting?

    - At which epoch does validation loss start increasing?

6. Evaluate model on test dataset

Bonus Question: Transfer Learning

7. Using a pretrained convolutional neural network such as MobileNetV2, perform transfer learning on the garbage classification dataset.

8. Load the pretrained model with:

    - include_top = False

- weights = 'imagenet'
- Freeze the pretrained layers.

9. Add the following layers on top:
   - GlobalAveragePooling layer
   - Dense output layer (Softmax, number of classes)
10. Train the model using the same parameters and evaluate the model on the test dataset.